

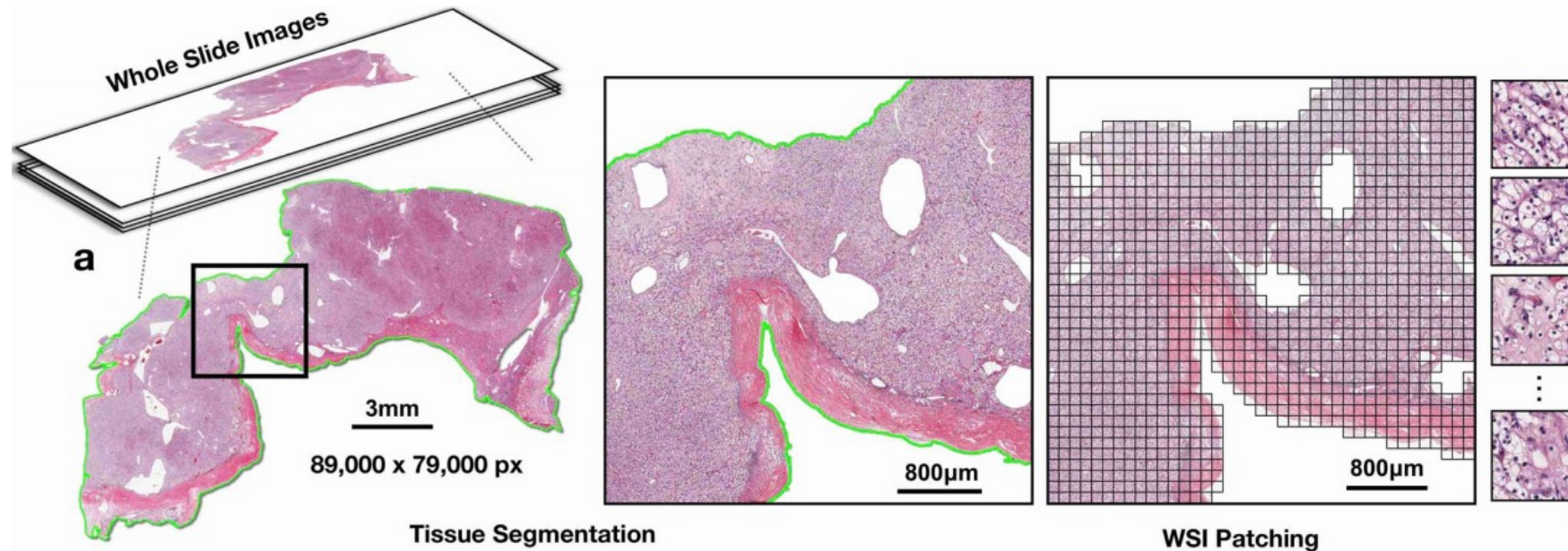
Memory Efficient Convolution Layer

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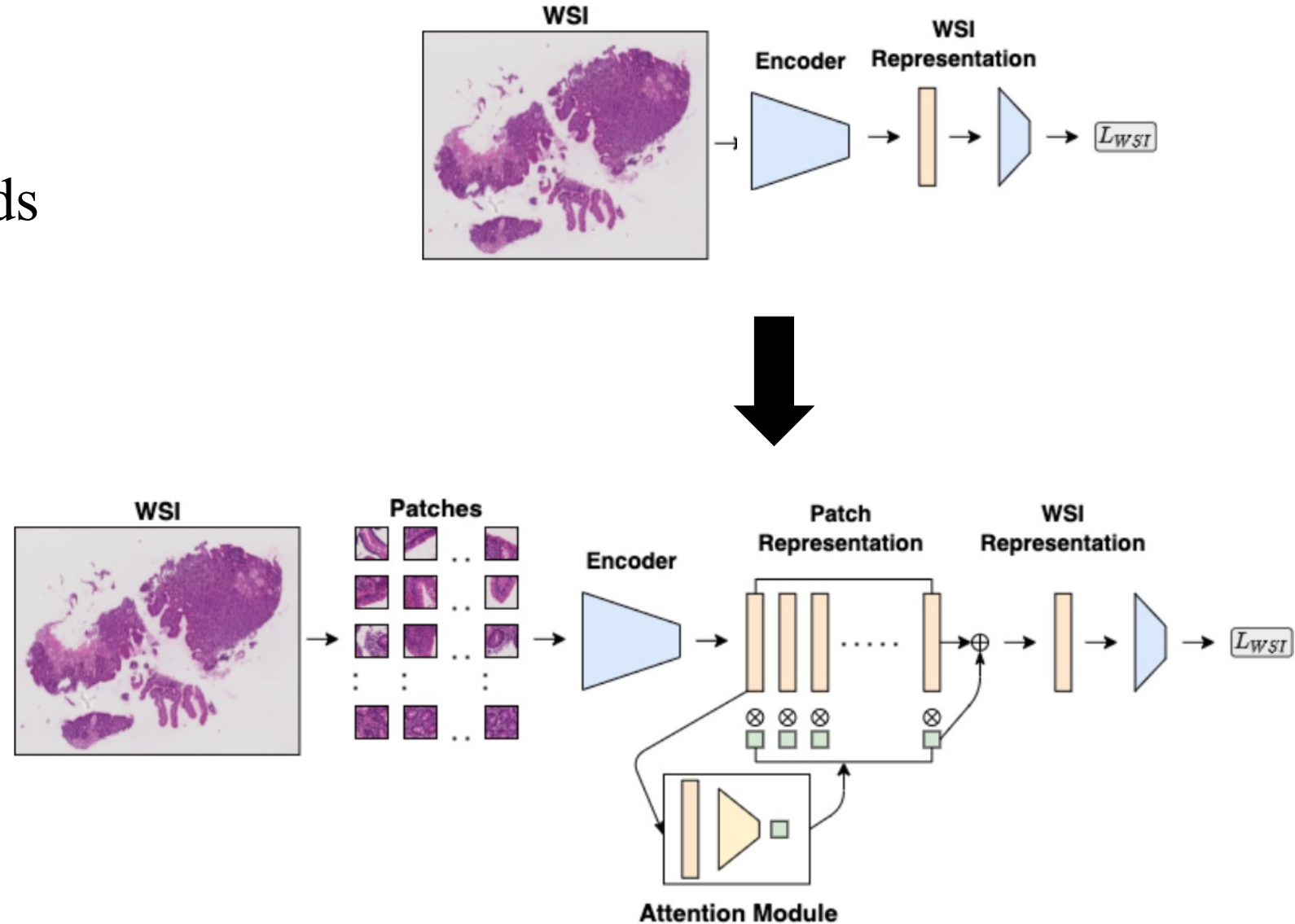
Introduction: WSI

- Whole Slide Imaging (WSI) in pathology involves extremely large images, posing significant challenges for analysis.

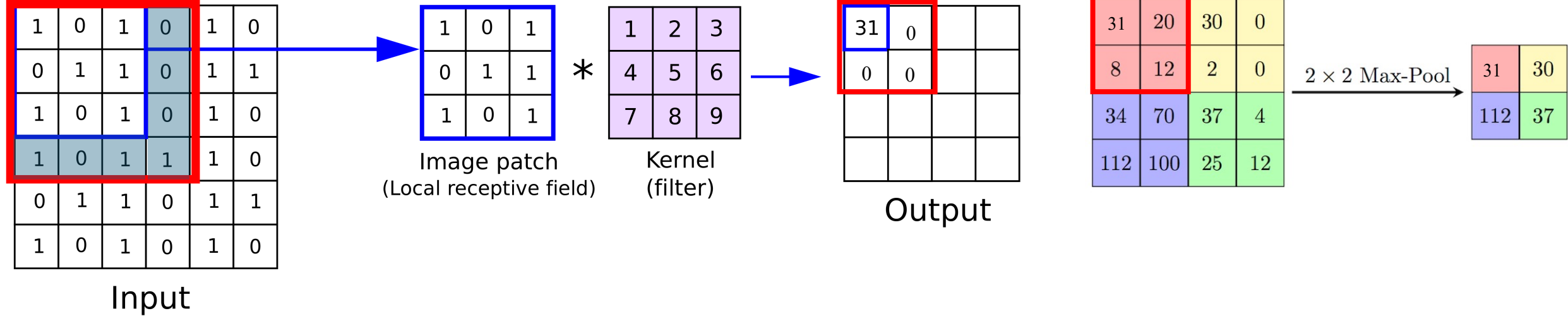


Introduction: patch-based

- Traditional approaches use patch-based methods and Multiple Instance Learning (MIL).



Introduction: memory



Method: Split the input

1	0	1	1	1	0	1	0	1	1	1	0
0	1	1	0	1	1	0	1	1	0	1	1
1	0	1	0	1	0	1	0	1	0	1	0
1	0	1	1	1	0	1	0	1	1	1	0
0	1	1	0	1	1	0	1	1	0	1	1
1	0	1	0	1	0	1	0	1	0	1	0
1	0	1	1	1	0	1	0	1	1	1	0
0	1	1	0	1	1	0	1	1	0	1	1
1	0	1	0	1	0	1	0	1	0	1	0
1	0	1	1	1	0	1	0	1	1	1	0
0	1	1	0	1	1	0	1	1	0	1	1
1	0	1	0	1	0	1	0	1	0	1	0



1	0	1	1	1	0	1	0	1	1	1	0
0	1	1	0	1	1	0	1	1	0	1	1
1	0	1	0	1	0	1	0	1	0	1	0
1	0	1	1	1	0	1	0	1	1	1	0

*

1	2	3
4	5	6
7	8	9

Kernel
(filter)

1	0	1	1	0	1	1	0	1	1
1	0	1	1	0	1	1	0	1	1

+

1	0	1	1	0	1	1	0	1	1
1	0	1	1	0	1	1	0	1	1

+

1	0	1	1	0	1	1	0	1	1
1	0	1	1	0	1	1	0	1	1

+

1	0	1	1	0	1	1	0	1	1
1	0	1	1	0	1	1	0	1	1

Method: Throw away

1	0	1	1	1	0	1	0	1	1	1	0
0	1	1	0	1	1	0	1	1	0	1	1
1	0	1	0	1	0	1	0	1	0	1	0
1	0	1	1	1	0	1	0	1	1	1	0
0	1	1	0	1	1	0	1	1	0	1	1
1	0	1	0	1	0	1	0	1	0	1	0
1	0	1	1	1	0	1	0	1	1	1	0
0	1	1	0	1	1	0	1	1	0	1	1
1	0	1	0	1	0	1	0	1	0	1	0
1	0	1	1	1	0	1	0	1	1	1	0
0	1	1	0	1	1	0	1	1	0	1	1
1	0	1	0	1	0	1	0	1	0	1	0

conv2d



1	0	1	1	0	1	1	0	1	1
1	0	1	1	0	1	1	0	1	1
+									
1	0	1	1	0	1	1	0	1	1
1	0	1	1	0	1	1	0	1	1
+									
1	0	1	1	0	1	1	0	1	1
1	0	1	1	0	1	1	0	1	1
+									
1	0	1	1	0	1	1	0	1	1
1	0	1	1	0	1	1	0	1	1

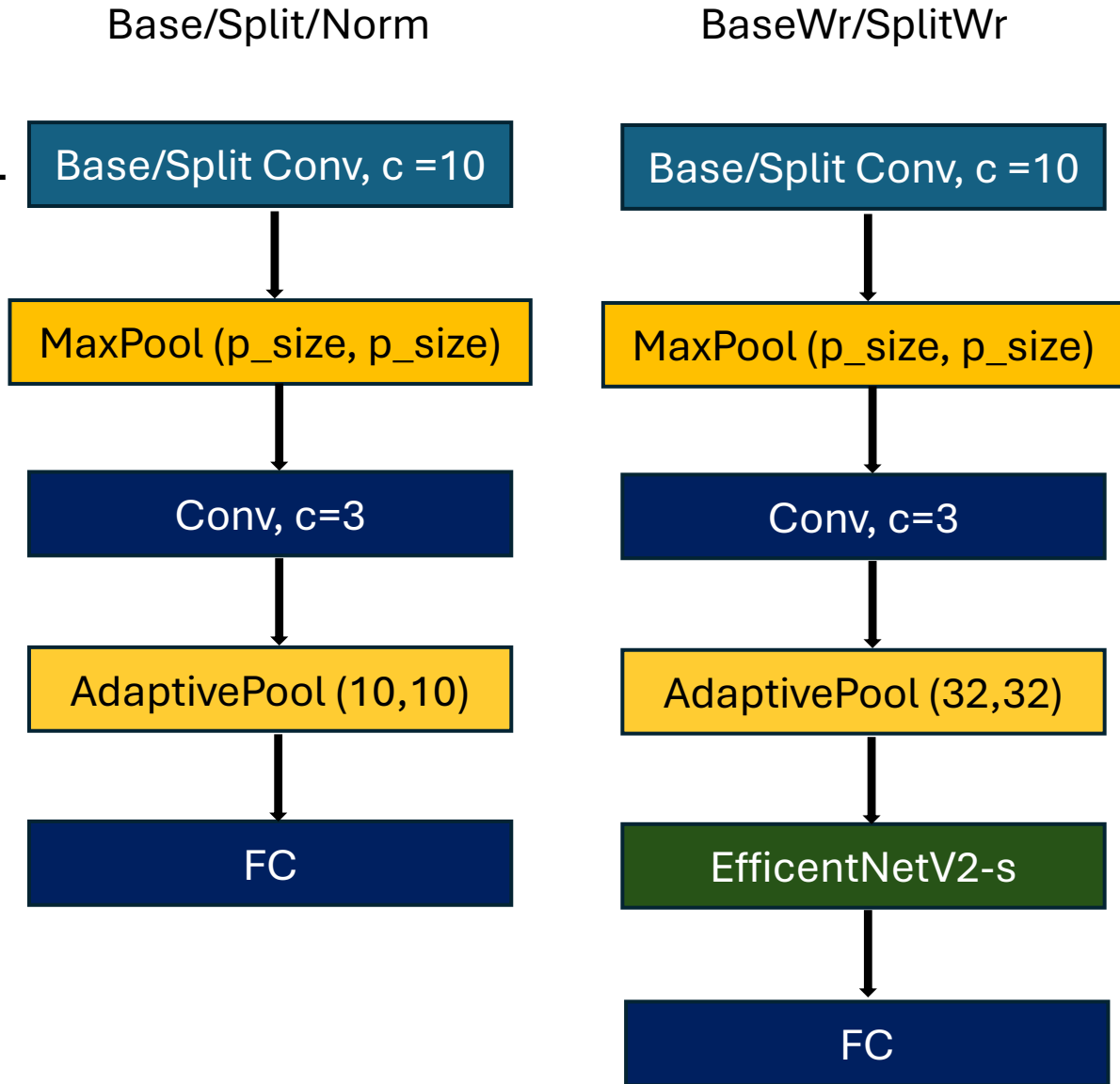
Max pooling



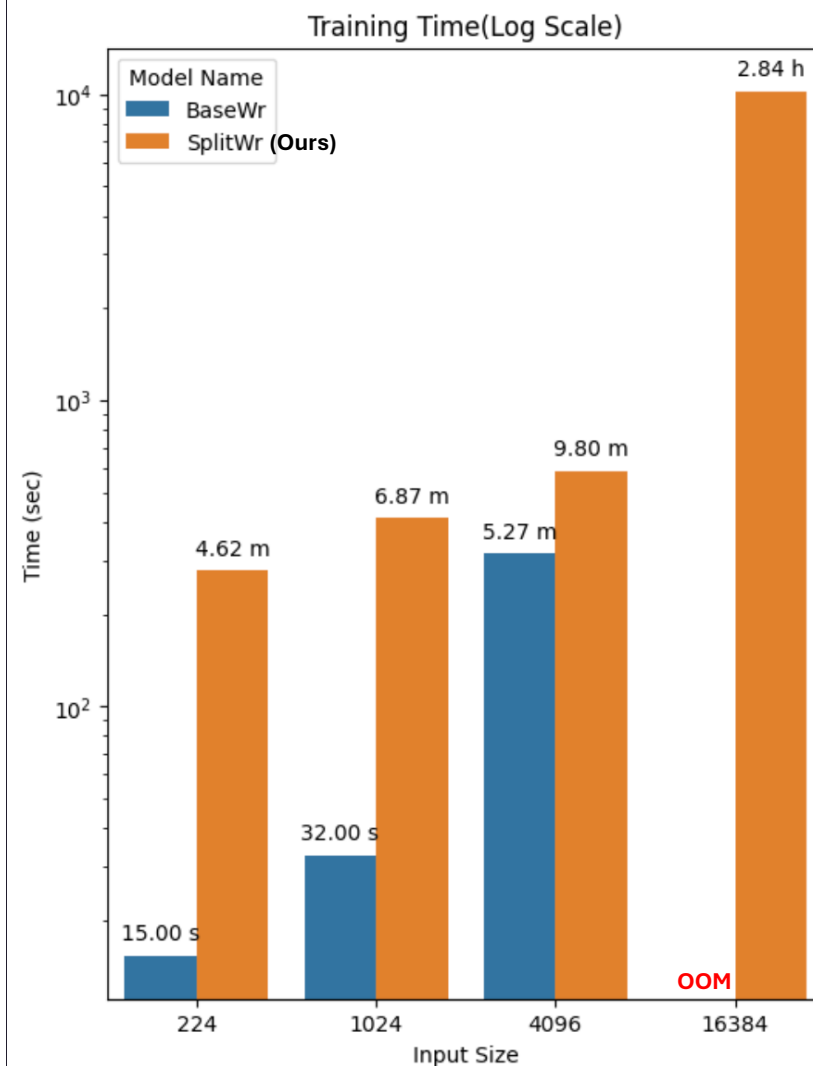
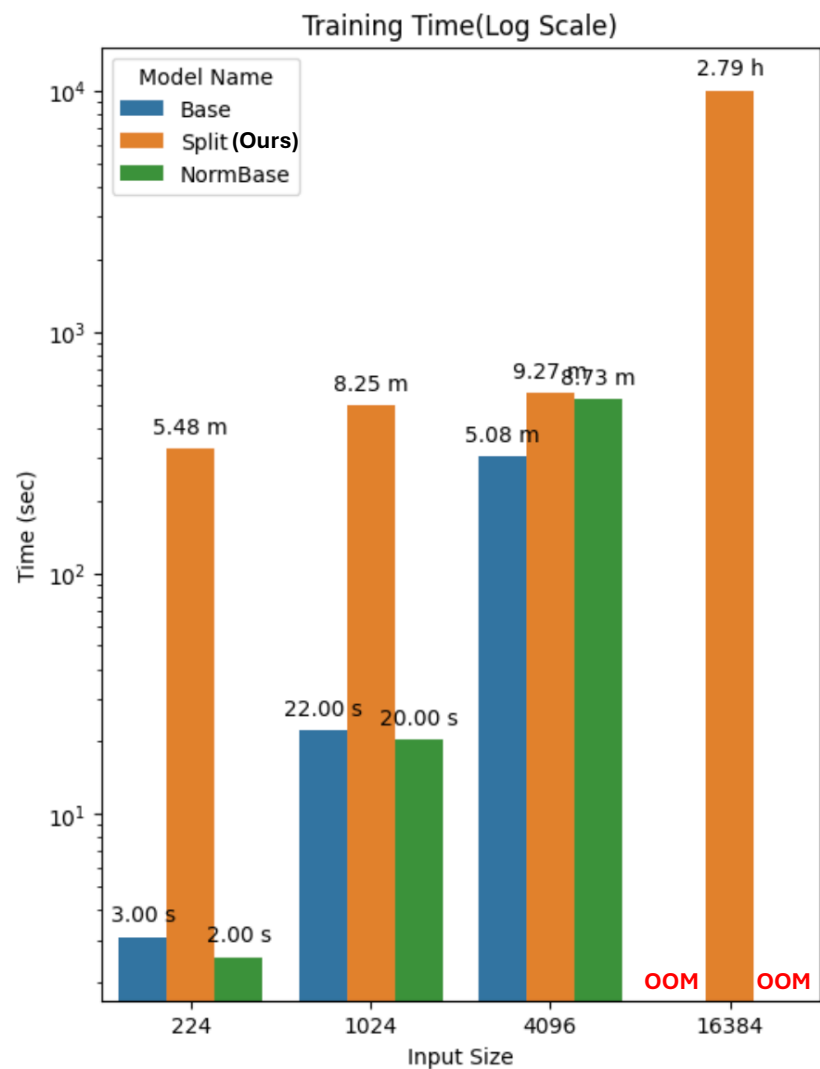
1	0	1	1
1	1	0	1
1	0	1	1
1	0	1	1

Experiment

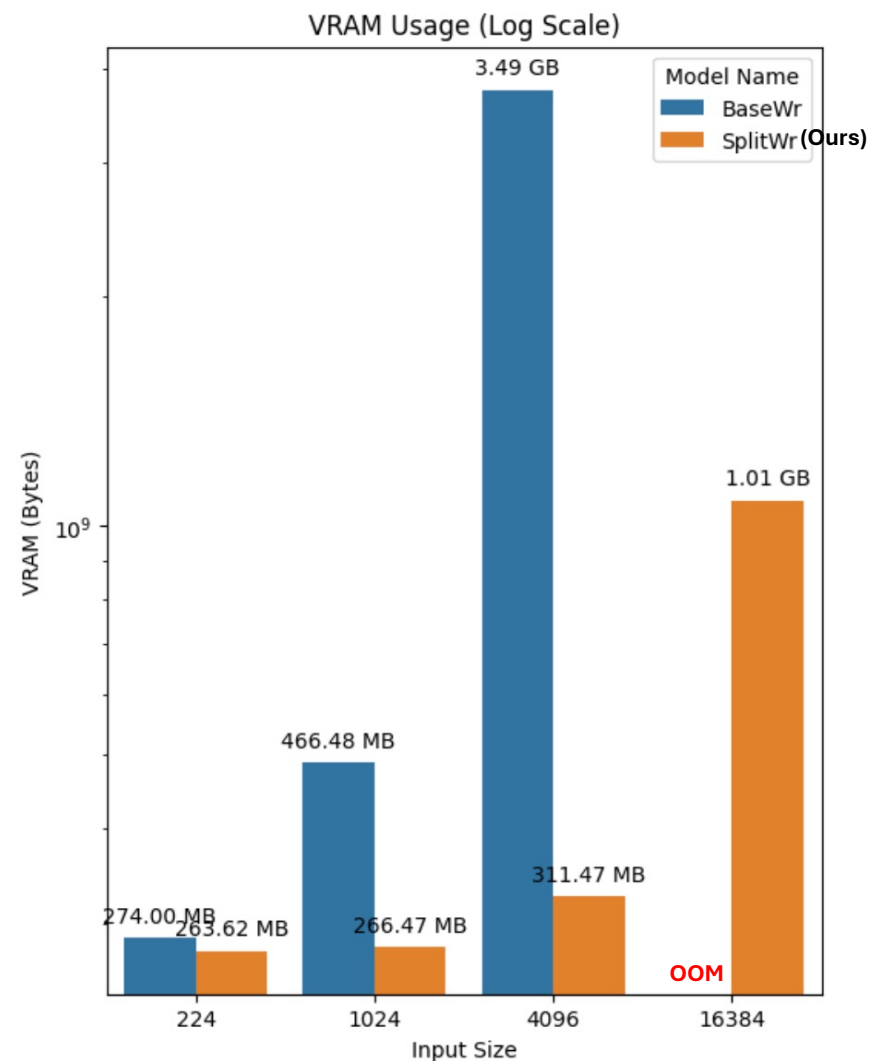
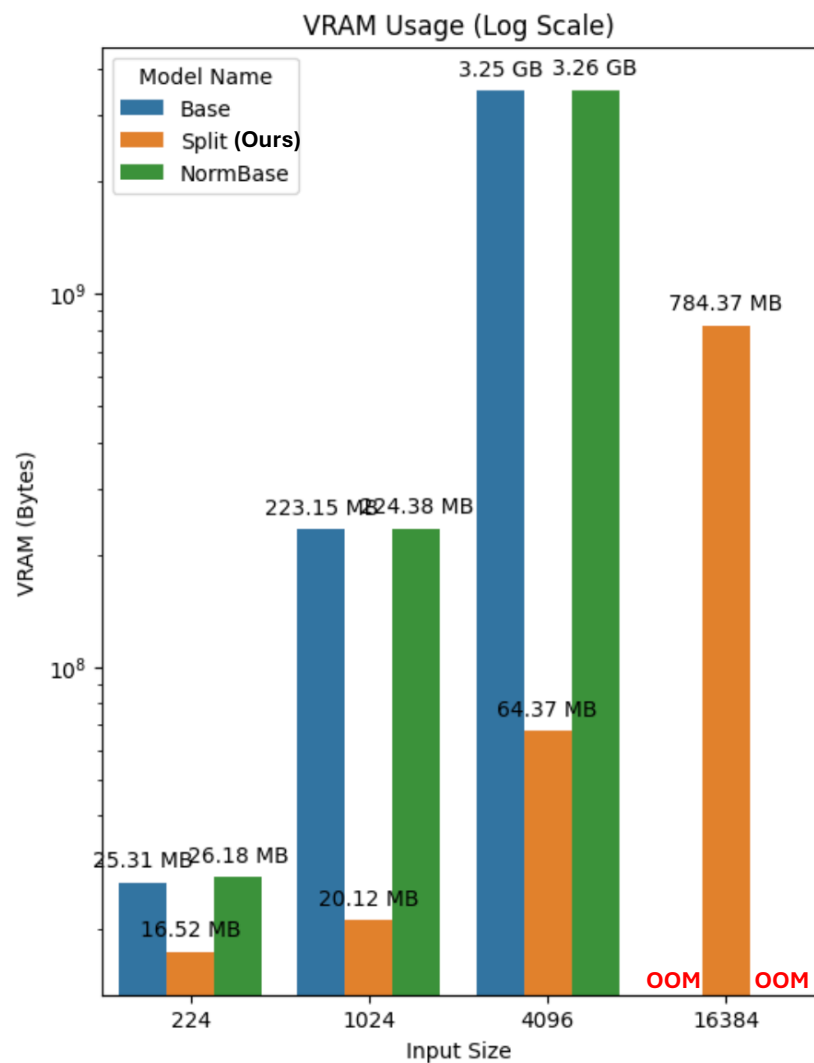
- Input size: 224, 1024, 4096, 16384
- Pooling size: 14, 62, 254, 1022
- Batch size: 4
- Epoch: 20



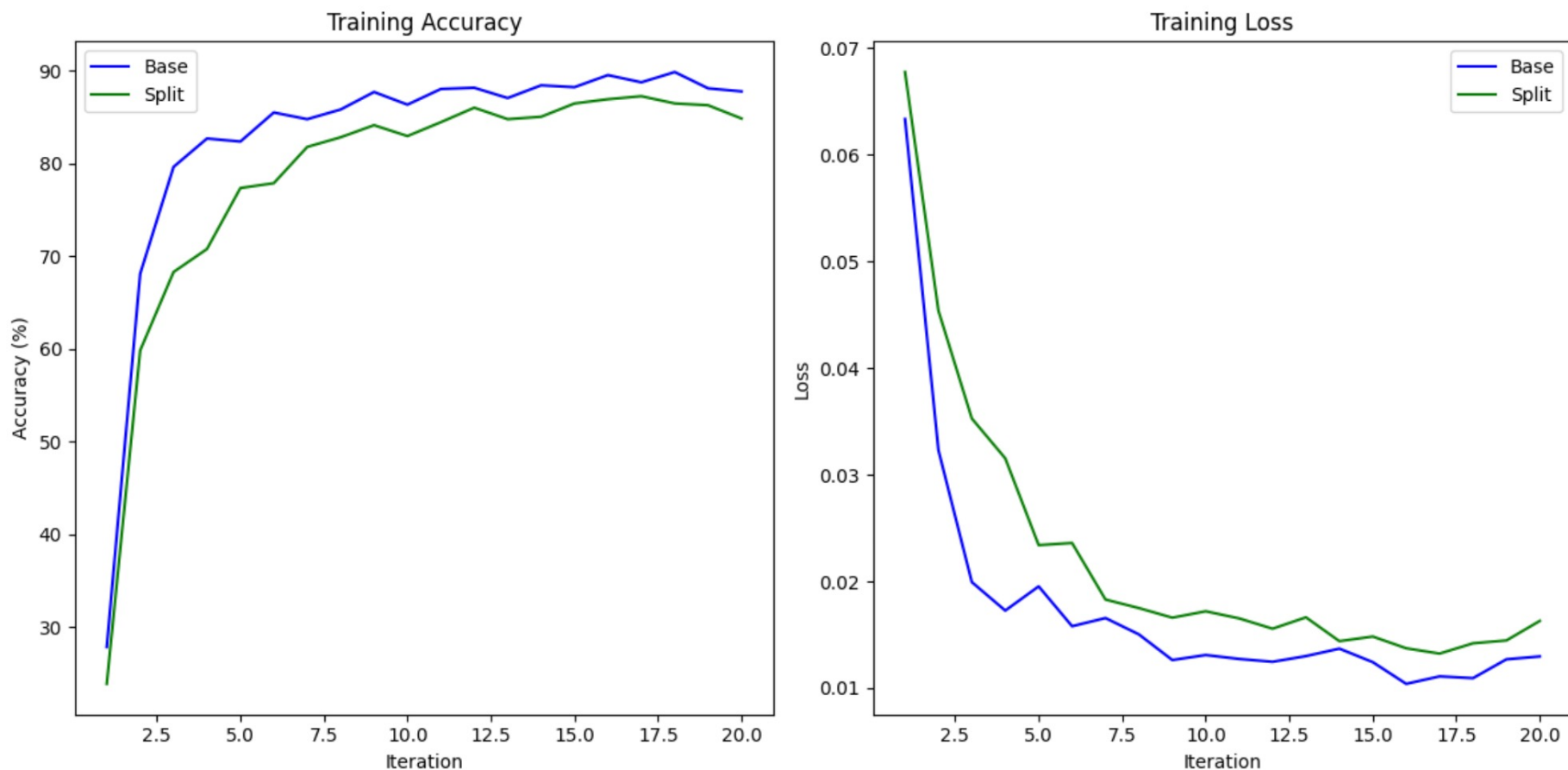
Experiment: Training Time



Experiment: VRAM Usage

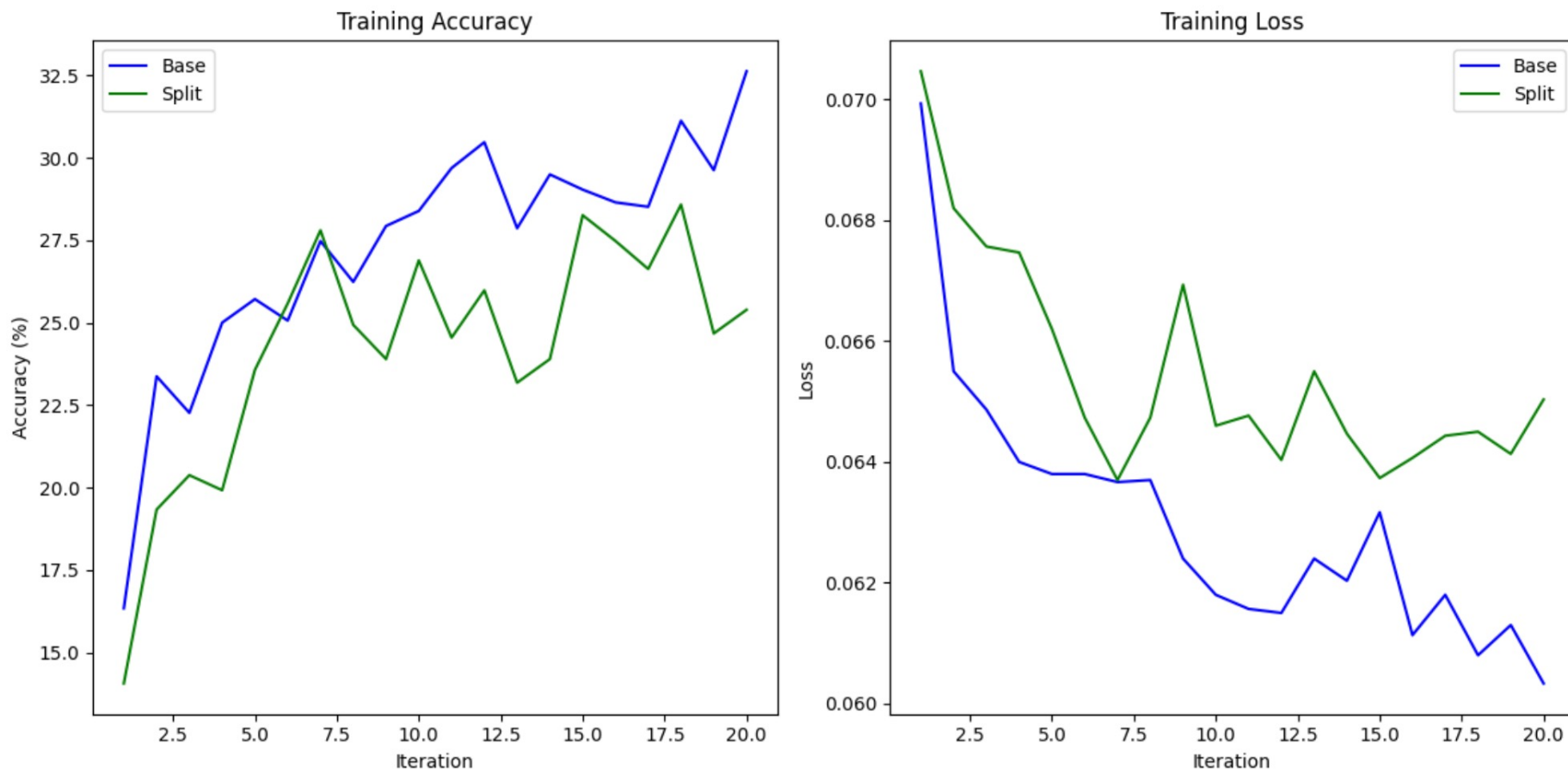


Experiment: MNIST



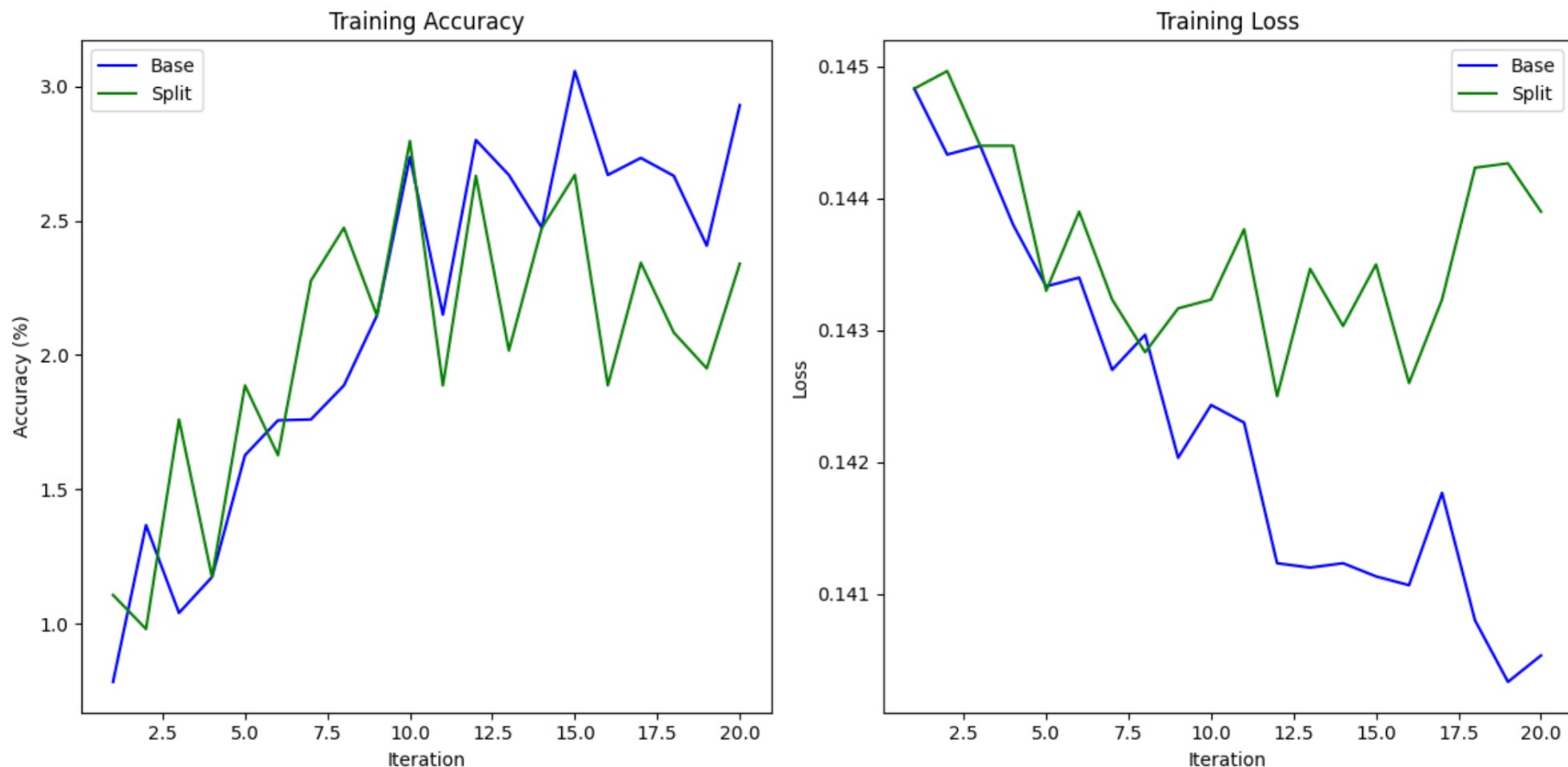
(base) Mean Test Accuracy: 90.365% Mean Test Loss: 0.008607048114451269
(split) Mean Test Accuracy: 91.146% Mean Test Loss: 0.008862371051994463

Experiment: cifar10



(base) Mean Test Accuracy: 34.375% Mean Test Loss: 0.05974315789838632
(split) Mean Test Accuracy: 29.948% Mean Test Loss: 0.06397256286193927

Experiment: food101



(base) Mean Test Accuracy: 2.604% Mean Test Loss: 0.14143195872505507
(split) Mean Test Accuracy: 1.823% Mean Test Loss: 0.14346491172909737

Conclusion & Problems

- Memory usage: Lower
- Performance: Suppose to be equivalent
- Training time: OK for massive image

Problems

- Semantics: Information loss
- Coding: Hard

Other

- Current work: Attention

